

FATIMA JINNAH WOMEN UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

COAL PROJECT REPORT



SUBMITTED TO :

MR MAJID SHAFIQUE

SUBMITTED BY:

**MAIMOONA BIBI (055)
MALAYKA IBRAR (058)
NATASHA MARYAM (068)
MALIHA ZAFAR (096)**

Project Title:

Home Security System

1. Introduction

Security is a primary concern in today's world, and a reliable home security system plays a crucial role in protecting our properties. This project presents a simple yet effective Home Security System that uses an Arduino Nano, a PIR motion sensor, and an RS232 communication setup. It is designed to detect unauthorized movement and respond instantly by sounding an alarm and lighting up an LED. Additionally, the system communicates motion detection status to a computer using MAX232 and RS232 to USB cable.

2. What the Project Does

The system continuously monitors the area using a PIR motion sensor. When motion is detected:

- A red LED turns on.
- A buzzer is triggered.
- A message is sent via RS232 to the computer.

The alarm remains active until a push button is pressed to acknowledge and stop it.

3. Real-life Applications

- Home intrusion detection
- Office or warehouse security

- Small shop anti-theft systems
 - Motion-triggered alert systems
 - Temporary security installations (events, exhibitions)
-

4. Component Connections (Table Format)

Component	Arduino Nano Pin	Description
PIR Sensor OUT	D2	Detects motion
Red LED (+)	D3	Indicates motion
Buzzer (+)	D4	Sounds alarm on motion
Push Button	D5	Turns off the alarm when pressed
MAX232 RX	D7 (TX via SoftSerial)	Sends serial data to PC
MAX232 TX	D6 (RX via SoftSerial)	(Not used, reserved)

5. Role of MAX232 and RS232 Cable

The MAX232 converter converts the 5V TTL logic level from the Arduino to the RS232 voltage levels required by the PC's serial port. The RS232 to USB cable allows communication between Arduino and the computer. This enables status messages to be displayed on a terminal like PuTTY or Arduino Serial Monitor, enhancing the project's usability for remote monitoring.

6. Code

```
#include <SoftwareSerial.h>

#define PIR_PIN    2
#define LED_PIN    3
#define BUZZER_PIN 4
#define BUTTON_PIN 5

SoftwareSerial mySerial(6, 7); // RX, TX

bool alarmActive = false;

void setup() {
  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT_PULLUP);

  digitalWrite(LED_PIN, LOW);
  digitalWrite(BUZZER_PIN, LOW);

  Serial.begin(2400);
  mySerial.begin(2400);

  Serial.println("System Ready");
  mySerial.println("System Ready");
}

void loop() {
  if (digitalRead(PIR_PIN) == HIGH && !alarmActive) {
    alarmActive = true;
    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
    Serial.println("Motion Detected!");
    mySerial.println("Motion Detected!");
  }

  if (alarmActive && digitalRead(BUTTON_PIN) == LOW) {
    alarmActive = false;
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    Serial.println("Alarm Stopped.");
    mySerial.println("Alarm Stopped.");
  }
}
```

```
    delay(300);  
  }  
}
```

7. Conclusion

This project demonstrates a reliable and responsive home security system using Arduino Nano and minimal hardware. With its real-time motion detection, alarm activation, and RS232 communication, it provides an affordable and efficient solution for small-scale security applications.

8. References

1. Arduino Official Documentation - <https://www.arduino.cc>
 2. MAX232 Datasheet - <https://www.ti.com/product/MAX232>
 3. RS232 Communication Basics - <https://learn.sparkfun.com/tutorials/serial-communication>
 4. PIR Sensor Usage Guide - <https://randomnerdtutorials.com/pir-motion-sensor-arduino-tutorial/>
-